instacart

Case Study

Instacart Agent Readiness – Product



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Where I think the work is

Finding products is the easy part for an agent. The hard part is the small calls a person makes while shopping: which oat milk to take when theirs is sold out, whether "2" means two bags or two kilos, whether a fifth order this month was actually wanted. Most of my answers come back to writing those calls down so an agent can read them, then giving it more freedom as the data supports it.

1 Why start with groceries?

The job repeats every week, customers already save their replacement choices, and a mistake usually means refunding one item.

Loyal users order 3.9×/month · 3 saved replacement options · top category people would let AI buy (28%)

2 The flashy board demo

Build a demo on top of the substitution feature from Q3. Scripted parts stay under 5 engineer-days.

ChatGPT's in-chat checkout was pulled after about 6 months; Walmart saw it convert about 3× worse

3 Checkout or substitutions?

Substitutions. It's the step where the agent has to make a call while a shopper waits, and agents can't read those preferences today.

Instacart's public agent tools: 2, none for replacements · my scoring 4.8 vs 1.8

4 "quantity: 2" read two ways

Mostly our API contract. Require a unit, check the total against what the agent expected, and send back what we understood.

Instacart itself replaced its bare quantity field with a measurements list in 2026

5 Orders per user per month

4 a month, at most one in any 7 days. A fifth order goes through after one tap from the user.

Loyal users average 3.9/month · 300 clean orders put the unwanted-order rate under 1% (95% confidence)

6 "Should we be worried?"

Some worry is fair. I'd show the client the main risks, what we've done about each, and the points where we'd pause.

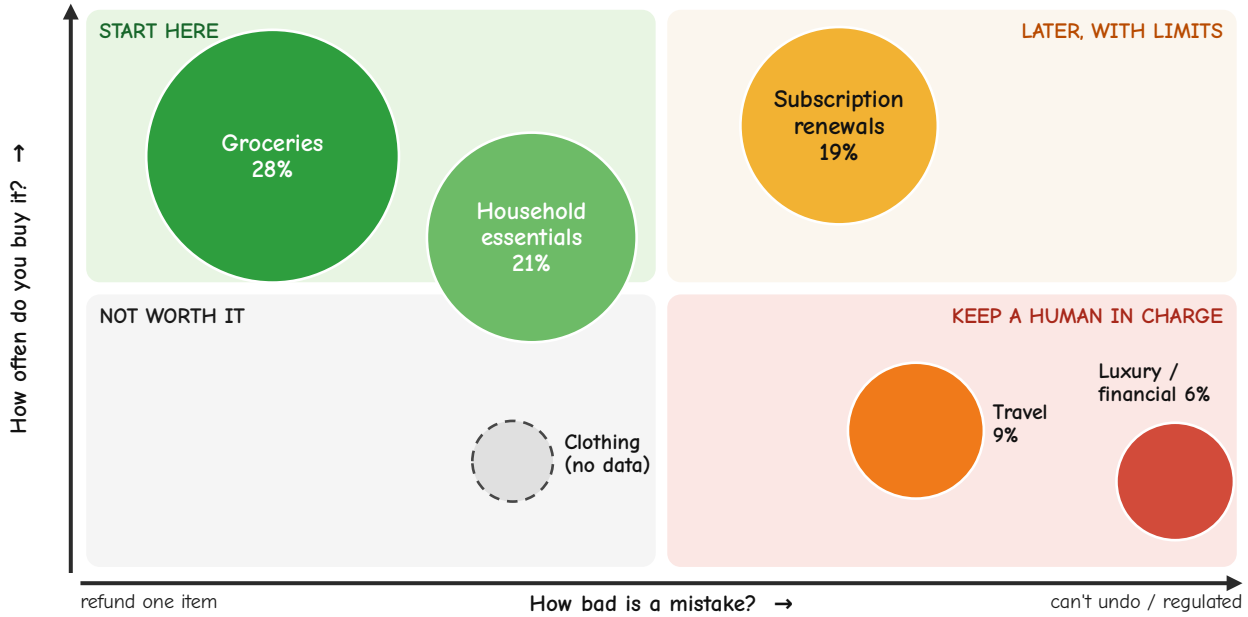
Gartner expects 40%+ of agent projects to be cancelled by 2027 · Instacart's own $60M FTC consent case

90.3M orders last quarter, +9% YoY · **$115** average order value · **1.6B** orders ever placed · **2B+** catalog items, 2,200 retailers · **3** AI platforms: ChatGPT, Claude, Gemini · **1** own agent: Clementine

Figures: Instacart Q2 2026 results [5]; Clementine launch coverage [4]; ChatGPT, Claude and Gemini integrations [1][2][3]. Sources are listed in the appendix.

Groceries are a good first job for an agent: people repeat the same shop, their preferences are already saved, and mistakes are cheap to fix

Which kinds of shopping could an agent take over first?



Bubble size is the share of US adults who'd let AI buy in that category [17]. ACI's survey ranks them the same way (groceries 41%, household 31%) [18]. Where each one sits on the axes is my judgment.

Stock-outs are where the agent earns its keep

About two-thirds of Instacart orders include meat or produce [5], and stock changes all day (10M+ inventory signals daily [4]). Typing the list is easy; handling the missing item the way the customer would is the real test, which is why I pick substitutions in Q3. I'd start with one opt-in job, "refill my usual basket within my rules", and track how many of those baskets arrive with no edits and no issues.

What makes grocery a safe place to start

IT REPEATS

3.9×

orders / month

Instacart's monthly customers order 2.1 times a month in their first year and 3.9 by year six [7]. US shoppers make 1.6 grocery trips a week [23], and Instacart's own assistant already handles "order my usuals" [4].

RULES EXIST

3

options on file

For each item, customers can already name up to three backups, let the shopper pick the best match, or ask for a refund. Approved replacements are saved for next time, with tags like same brand or gluten-free [10].

EASY TO UNDO

95%

happy with a replacement

Orders with a replacement get 95% satisfaction, and replacements per order are at their lowest since 2022 [6]. A wrong item gets refunded; a wrong flight usually can't be.

MEASURABLE

1.6B

orders ever

2B+ catalog items across 2,200 retailers [4], and order quality has improved 16 quarters in a row [5]. Agent orders can be compared with normal orders from the first week.

Why now

Gemini can already run Android apps on its own [24], and a US appeals court ruled that an agent acting for a user counts as the user [16]. Agents are coming either way.

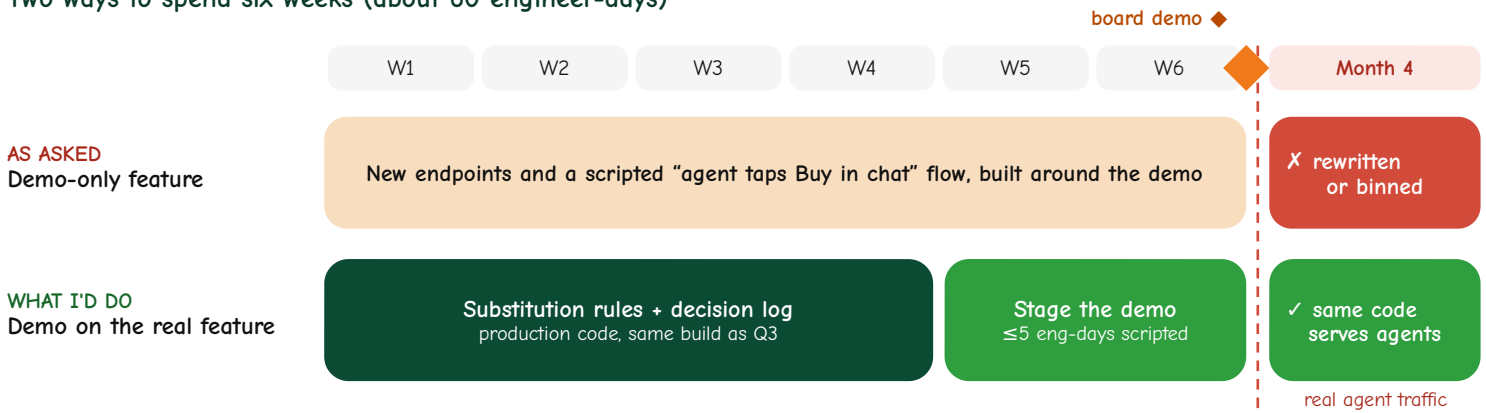
The catch

Only 22% of people who use AI to shop trust an agent to decide purchases [19]. So I'd make it opt-in, limited to the usual basket, and start with drafts (Q5).

[4] Clementine launch · [5] Instacart Q2 2026 results · [6] Q2 2026 earnings call · [7] Instacart S-1 / Q3 2023 call · [10] Instacart Help Center · [16] Ninth Circuit, Amazon v. Perplexity · [17] Supermarket News · [18] ACI Worldwide · [19] FoodNavigator · [23] FMI Trends · [24] 9to5Google

I'd give the board a demo, but build it on the substitution feature we need anyway, so it's still in use by month 4

Two ways to spend six weeks (about 60 engineer-days)



Before building anything, I'd sort each demo piece by what happens to it afterwards

KEEP · real, stays

~48 eng-days · 80%

- Substitution rules API
- Decision log (what it saw, the rule, the choice)
- Timeout and refund fallback
- Agent tools (MCP)

WRAP · scripted, calls the real API

~7 eng-days · 12%

- Fake stock-out in the inventory feed
- Demo family and usual basket
- Presenter screen on real responses

THROW · only for the stage

≤5 eng-days · 8% (cap)

- Stage animation and narration
- Hard-coded timing for the big moment
- Anything that would change the real API is out

What the board would see (only the stock-out is scripted)



Where I'd give way, and where I wouldn't

- Screens, narration and demo data are easy to change later, so I'd build them quickly and cap the time.
- The API contract, data model and public agent tools are hard to undo. I wouldn't change any of them just for the demo, and that's the one point I'd hold firm on.
- The demo should be a thin version of the real feature that the team keeps building on, so nothing needs rebuilding after the board meeting.

Why I think the flashy version won't last

- Instacart was first to launch checkout inside ChatGPT [1]. OpenAI retired that checkout about six months later and sent shoppers back to merchants' own sites [12].
- Walmart said purchases made inside ChatGPT converted at about a third of the rate of its own site, and called the experience "unsatisfying" [11].
- Gartner expects over 40% of agent projects to be cancelled by 2027, mostly over cost, unclear value or weak risk controls [13].

How I'd put it to the team

"Let's show the board something that's still running in month four. Pick the moment you want on stage, I'll build it on the real feature, and we'll say which parts are scripted." If they still want the original, I'd write my concern down and go with their call, and only escalate if it meant changing the real API.

[1] Instacart × ChatGPT launch · [11] Search Engine Land, Walmart ChatGPT checkout · [12] CNBC, OpenAI revamps ChatGPT shopping · [13] Gartner. The effort split and 5-day cap are my proposals, to agree with the team.

Substitutions first. That's where an agent gets stuck in the middle of an order, and checkout already has workable options

Where an agent-run order gets stuck



How I scored the two options (weights and scores are my judgment)

What matters	Weight	Checkout	Substitutions
Agent has to make a judgment call	25%	2	5
A person is waiting on the answer	20%	1	5
Not already covered by payment standards	15%	1	5
Needs Instacart's own data (1.6B orders)	15%	2	5
Cost of shipping it late	15%	2	4
Fits 2 engineers × 4 weeks (no PCI / fraud)	10%	3	4
Weighted total (out of 5)		**1.8**	**4.8**

Draft rule an agent reads and sets for each item in the usual basket

```
{ "item": "upc:0001111041600", "mode": "specific_backups",   // or best_match / refund
  "backups": ["upc:…", "upc:…"], "max_price_change_pct": 10, "must_match": ["dairy_free"],
  "if_no_answer_after_s": 300, "fallback": "refund", "set_by": "agent:phone-assistant" }
```

Why checkout can wait

- Standard ways for agents to pay already exist or are on the way: ACP from OpenAI and Stripe, UCP and AP2 from Google, and Visa and Mastercard agent tokens with spending limits [21][22].
- Checkout has moved back to merchants. ChatGPT's in-chat checkout was retired [12], and the Claude and Gemini integrations send people to Instacart to pay [2][3].
- Proper checkout work means payments, fraud and chargebacks, which is more than two engineers can do safely in four weeks.

Why substitutions

- Instacart's public agent tools can create recipes and shopping lists [9], but nothing reads or sets replacement preferences.
- When an item is out, a person answers the shopper's message in the aisle. An agent can only do that if the preference is written down somewhere it can read.
- Instacart already has the options [10] and a new replacement model that takes diet into account [6]; the work is making them readable by agents.

Plan for the 4 weeks

- **W1** Audit what exists; design the rule format and limits
- **W2** Agents can read and set rules (API and agent tools)
- **W3** Log every decision with a reason; no answer means rule, else refund
- **W4** Test: all backups out, +40% price, allergy, two agents clash

If agents had no way at all to pay or hand the customer over, I'd do checkout first instead.

[2] Claude integration · [3] Gemini integration · [5] Q2 2026 results · [6] Q2 2026 call · [8] developer docs · [9] Instacart MCP docs · [10] Help Center · [12] CNBC · [21] Google Developers, UCP/AP2 · [22] PYMNTS, Visa and Mastercard agent tokens

Most of the ambiguity comes from our API contract, so that's where I'd fix it

Five places "2" can lose its meaning, most likely first

Agent A → 2 units (any pack size)

SAME REQUEST

```
{"name": "wheat flour", "quantity": 2}
```

Agent B → 2 × 1 kg packs = 2 kg

Layer	What goes wrong	Likelihood	Evidence
1 API / agent contract	A bare number with an optional unit, so "2" has no kind (item, pack, weight).	HIGHEST	Instacart replaced quantity + unit on another endpoint in 2026 [8]
2 Server resolver	Given several measures, the server quietly picks the "best" one and the agent never sees which.	HIGH	Same changelog: "Instacart can choose the best one" [8]
3 Catalog	"Wheat flour 1 kg" could be a pack, loose weight or a multipack, and it varies by store.	HIGH	2B+ items, 2,200 retailers [4]
4 Tool description	The tool text never says what quantity counts, so each platform guesses.	MEDIUM	Public tools: create-recipe, create-shopping-list [9]
5 Agent parsing	Each model turns "2 bags of atta" into data its own way. We can't control this part.	MEDIUM	My inference

Today

```
{
  "name": "wheat flour",
  "quantity": 2
}


// 2 what? item? pack?
// kg? server guesses
```

Proposed request

```
{ "product": {"upc": "8901725…"},
  "measure": {
    "kind": "pack", "count": 2,
    "pack_size": {"value":1,"unit":"kg"}},
  "expect_total": {"value":2,"unit":"kg"},
  "on_mismatch": "reject" }
```

Instacart's reply

```
{ "resolved": "2 × Atta 1 kg pack",
  "total": {"value":2.0,"unit":"kg"},
  "matched_by": "barcode",
  "check": "MATCH",
  "status": "OK" }
// shown to user + saved in log
```

First step: compare the two requests

Replay both requests from the logs. Are they identical?

- Yes: our side, resolver or catalog (layers 2–3)
- No: their side, tool text or parsing (layers 4–5)

Changes I'd make to the contract

1. Replace the bare number with a typed measure: item, pack, weight or volume. The 2026 MCP spec supports this kind of either/or schema [20].
2. Identify products by barcode. Instacart already prefers barcodes when they're sent [8].
3. Have the agent state the total it expects, and reject the order if ours differs.
4. Reply with what we understood ("2 × 1 kg pack = 2 kg") so a mismatch shows up before shopping starts.
5. If it's still unclear, ask the user through the protocol's form feature [20].

Keeping it fixed

A shared list of about 100 tricky products (multipacks, loose produce, dual units) that every agent partner runs before release. I'd track the read-back mismatch rate and add a "wrong quantity" support tag. The new format runs alongside the old one for 90 days before agents lose the bare field.

[4] Clementine launch (catalog size) · [8] Instacart developer changelog and shopping-list docs · [9] Instacart MCP docs · [20] Model Context Protocol spec, 2026 (tool schemas, asking the user). Test-list size and notice period are my proposals.

I'd start at 4 agent orders per user per month, with no more than one in any 7 days

4 orders a month without asking (launch limit)
one per rolling 7 days · a 5th order goes through after one tap · within the user's spend limit · opt-in

How a user moves from drafts to ordering on its own

R0 DRAFT
Agent builds the basket; the user presses Order.

Learn: edit rate
Move up after opting in and 4 drafts with under 10% of lines edited

R1 ORDER + HOLD
Agent orders, with a 60-minute cancel window before shopping starts.

Learn: cancel rate
Move up after 300 orders with no "didn't mean to" (under 1% at 95%)

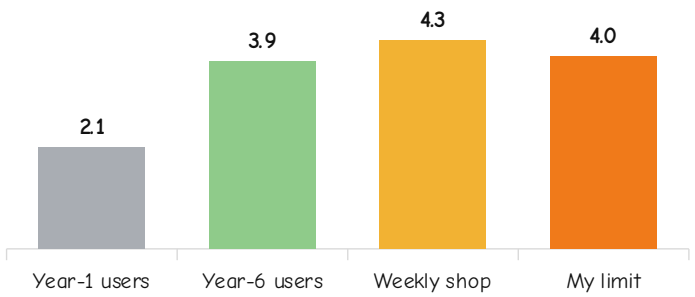
R2 ON ITS OWN
4 a month, one a week, inside the spend limit and substitution rules (Q3).

Learn: problems vs normal orders
Move up after 90 days at or below normal orders

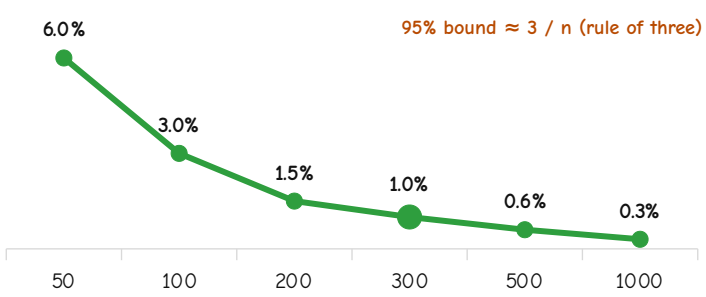
R3 PERSONAL
Limit set to the user's own usual pace + 1 (max 8), which they can raise.

Learn: long-run trust
A tripwire sends anyone back to R0

Orders per month, and where 4 sits [7]



Worst-case unwanted-order rate after n clean agent orders



Why 4

- Instacart's most loyal customers average 3.9 orders a month, first-year customers 2.1 [7]. Most people won't reach 4, so reaching it is worth a check.
- Months with five weekly slots come round about four times a year. The extra tap works as a regular re-consent, which matters after the FTC's 2025 order on consent for recurring charges [14].
- Order 5 still ships after the tap, so very few sales are lost.
- An unwanted order costs about $115 [5], mostly fresh food that gets refunded or thrown away. A tap costs the user a couple of seconds.
- The weekly rule matters most for bugs. With only a monthly cap, a looping agent could place 4 orders in an afternoon; the weekly rule stops it at the second.
- At 5, weekly shoppers would never re-confirm. At 10 (2.5× even loyal customers), a 10th order is more likely a bug than a need.

Other limits on every order

Spend	within the user's limit (default about 1.25× their usual basket)
New items	anything never bought before needs a tap
Substitutions	only inside the Q3 rules
Payment	limit also set on the card token [22]

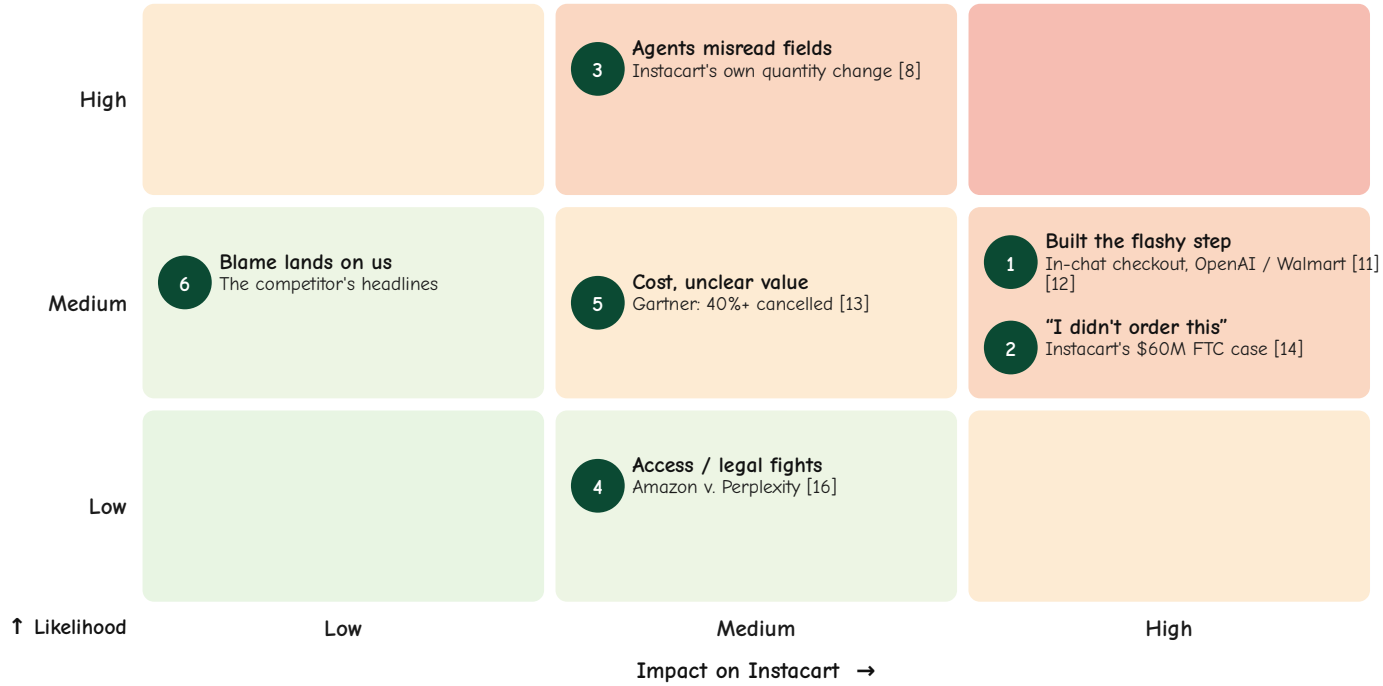
When I'd pull back (proposed)

"Didn't mean to order" above 0.5% of agent orders over 30 days sends that group back to R0. Cancels above 5%, or more problems than normal orders, pause promotions.

[5] Instacart Q2 2026 results ($115 average order) · [7] Instacart S-1 / Q3 2023 call (2017–22 cohorts) · [14] FTC v. Instacart settlement · [22] PYMNTS, Visa and Mastercard agent tokens. All thresholds are starting values to tune.

The risks are real, but they're ones we can see coming, and we'd agree up front on when to slow down

How projects like this usually fail, and where ours stands after the work in Q2–Q5



What I'd say to them

"It's a fair worry. From the outside we can't tell why their project stopped, so I won't guess. What I can show you is where projects like this usually go wrong, what we've already done about each of those, and three signals we'd act on. If any of them trips, I'll tell you straight away and bring the data."

Three signals we'd agree on in week 2

1. Agent orders have more problems than normal orders, beyond an agreed margin: no one moves up the ladder.
2. "Didn't mean to order" complaints pass the agreed rate: that group goes back to drafts.
3. A quantity mismatch reaches a shopper: we stop releases until it's fixed.

What's already in place for each one

1. Substitutions first, demo built on top (Q2, Q3)
2. Drafts before autonomy, plus a monthly limit (Q5)
3. Required units and read-back (Q4)
4. Agents come in through official tools with login scopes
5. Four-week slices, each with a written definition of done
6. A shared log of assumptions and who decided what

What we don't know yet

How often agent orders are unwanted, and how many people want this at all: only 22% of AI shoppers trust an agent to buy [19]. About four weeks of draft orders would give us the first real numbers.

[8] developer changelog · [11] Search Engine Land · [12] CNBC · [13] Gartner · [14] FTC · [16] Cooley, Ninth Circuit ruling · [19] FoodNavigator

Appendix

Sources. The numbers in square brackets on each slide point to this list.

1. **Instacart** · 2025 · company.instacart.com/pressreleases
   Instacart app launches in OpenAI ChatGPT, with Instant Checkout
2. **Supermarket News** · 2026 · supermarketnews.com
   Instacart connects with AI platform Claude
3. **Instacart** · 2026 · company.instacart.com/updates
   Instacart brings agentic grocery shopping to Gemini
4. **TechCrunch / PYMNTS** · 2026 · techcrunch.com
   Instacart launches AI shopping assistant Clementine
5. **Instacart** · 2026 · investors.instacart.com
   Second quarter 2026 financial results and prepared remarks
6. **Instacart** · 2026 · investors.instacart.com
   Q2 2026 earnings call: replacement model, 95% satisfaction
7. **Maplebear (Instacart)** · 2023 · sec.gov
   S-1 filing and Q3 2023 earnings call: order frequency by cohort
8. **Instacart Docs** · 2026 · docs.instacart.com/developer_platform_api
   Developer Platform changelog and create-shopping-list page
9. **Instacart Docs** · 2026 · docs.instacart.com/developer_platform_api
   Connect your AI agent to Instacart with MCP
10. **Instacart Help Center** · instacart.com/help
    Adding item and replacement instructions
11. **Search Engine Land** · 2026 · searchengineland.com
    Walmart: ChatGPT checkout converted 3× worse than website
12. **CNBC** · 2026 · cnbc.com
    OpenAI revamps shopping experience in ChatGPT after Instant Checkout
13. **Gartner** · 2025 · gartner.com/en/newsroom
    Over 40% of agentic AI projects will be cancelled by end of 2027
14. **FTC** · 2025 · ftc.gov
    Instacart to pay $60 million in consumer refunds
15. **Consumer Reports** · 2025 · consumerreports.org
    Instacart stops AI pricing tests
16. **Cooley** · 2026 · cooley.com
    Ninth Circuit rules on AI agent access (Amazon v. Perplexity)
17. **Supermarket News** · 2026 · supermarketnews.com
    AI agents gaining the most trust with grocery shoppers
18. **Retail Dive / ACI Worldwide** · 2026 · retaildive.com
    Consumers trust AI to advise, but don't consent to it purchasing
19. **FoodNavigator** · 2026 · foodnavigator.com
    Grocery shoppers use AI, but trust in AI buying agents lags
20. **Model Context Protocol** · 2026 · modelcontextprotocol.io
    Specification, 2026-07-28 release: tools and elicitation
21. **Google Developers Blog** · 2026 · developers.googleblog.com
    Under the hood: Universal Commerce Protocol (with AP2)
22. **PYMNTS** · 2026 · pymnts.com
    Visa and Mastercard put tokens in charge of AI commerce
23. **FMI** · 2026 · fmi.org
    U.S. Grocery Shopper Trends 2026
24. **9to5Google** · 2026 · 9to5google.com
    Gemini Android app automation rolls out
25. **PYMNTS** · 2026 · pymnts.com
    Anthropic debuts commerce agent blueprint with Visa and Mastercard

A note on limits: everything here comes from public sources. Cohort data covers 2017–22 customers and isn't agent-specific, and surveys measure trust differently. The thresholds I propose (4 a month, 0.5%, 5 engineer-days, 300 orders) are starting points I'd expect to change once real data comes in.